

Protective effects and mechanisms of sirtuins in the nervous system

Silent information regulator two proteins (sirtuins or SIRTs) are a group of histone deacetylases whose activities are dependent on and regulated by nicotinamide adenine dinucleotide (NAD⁺). They suppress genome-wide transcription, yet upregulate a select set of proteins related to energy metabolism and pro-survival mechanisms, and therefore play a key role in the longevity effects elicited by calorie restriction. Recently, a neuroprotective effect of sirtuins has been reported for both acute and chronic neurological diseases. The focus of this review is to summarize the latest progress regarding the protective effects of sirtuins, with a focus on SIRT1. We first introduce the distribution of sirtuins in the brain and how their expression and activity are regulated. We then highlight their protective effects against common neurological disorders, such as cerebral ischemia, axonal injury, Alzheimer's disease, Parkinson's disease, amyotrophic lateral sclerosis, and multiple sclerosis. Finally, we analyze the mechanisms underlying sirtuin-mediated neuroprotection, centering on their non-histone substrates such as DNA repair enzymes, protein kinases, transcription factors, and coactivators. Collectively, the information compiled here will serve as a comprehensive reference for the actions of sirtuins in the nervous system to date, and will hopefully help to design further experimental research and expand sirtuins as therapeutic targets in the future.